MICROSAR Ethernet Transceiver Driver

Technical Reference

Marvell 88Q2112/88Q2110

Version 3.1.0

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Status	Released

Document Information

History

Author	Date	Version	Remarks
David Röder	2017-07-10	1.00.00	Creation of document
Mark Harsch Martin Heiker	2018-09-03 2018-09-27	2.00.00	▶ 88Q2112STORYC-6361: Perform maintenance and reviews to achieve proc 3.0 QM release: TechRef ▶ Add section with limitations of Cable Diagnostics
Martin Heiker	2019-03-13	3.00.00	▶ Support of 1000Base-T1 SQI calculation
Martin Heiker	2019-03-13	3.00.01	▶ Fixed formulation error in caution section of 1000Base-T1 Mode SQI
Martin Heiker	2020-03-20	3.01.00	▶ Add section about Marvell's initialization sequence and errata bug fixes

Table 1-1 History of the document

Reference Documents

No.	Title	Version
[1]	TechnicalReference_EthTrcv_Core	see delivery
[2]	Marvell 88Q2110/88Q2112 Datasheet Doc. No. MV-S110850-00	Rev. A, March 23, 2017, Draft
[3]	Marvell SQI Marvell Automotive PHYs	January 2019
[4]	Marvell 88Q2110/88Q2112 Datasheet Doc. No. MV-S110850-00	Rev. G June 19, 2019
[5]	Marvell 88Q2110/88Q2112 Software Initialization Guide Doc. No. MV-S400460-00	Rev. -- October 32, 2019
[6]	Marvell Release Notes – 88Q2112 Rev A1 Doc. No. MV-S302907-00	Rev. A February 20, 2019
[7]	Marvell Release Notes – 88Q2112 Rev Z1 Doc. No. MV-S302776-00	Rev. -- August 24, 2017
[8]	Marvell Release Notes – 88Q2110/88Q2112 Rev A2 Doc. No. MV-S302915-00	Rev. A May 1, 2019

Table 1-2 Reference documents

Scope of the Document

This complement to the technical reference [1] describes the derivative dependent aspects of the Ethernet Transceiver Driver basis software. Please refer to your Release Notes to get a detailed description of the platform (host, compiler, and transceiver) your Vector Ethernet Bundle has been configured for.



Please note
We have configured the programs in accordance with your specifications in the questionnaire. Whereas the programs do support other configurations than the one specified in your questionnaire, Vector´s release of the programs delivered to your company is expressly restricted to the configuration you have specified in the questionnaire.



Note
This document describes aspects that apply for Ethernet Transceiver Driver modules provided by Vector, which are only used for Marvell 88Q2112 transceivers. For this reason, the corresponding infixes of APIs, macros, parameters etc. are replaced by 88Q2112.

The derivative independent aspects of the Ethernet Transceiver Driver are discussed in [1], where the above-mentioned infix is replaced by the markers <Derivative> and <DERIVATIVE>.

Contents

1 Component History 6

2 Introduction 7

3 Hardware Overview 8

4 Derivative Specific Functional Description 9

4.1 Overview 9

4.2 Limitations to supported Features and Functions 9

4.2.1 PHY Test Modes 9

4.2.2 PHY Loopback Modes 10

4.2.3 PHY TX Modes 10

4.2.4 PHY Cable Diagnostics 10

4.3 Marvell Specific Implementation 11

4.3.1 Supported hardware revisions 11

4.3.2 Initialization Sequence 11

4.3.3 1000Base-T1 Mode SQI 12

4.3.4 PAM2/PAM3 Link Training 13

4.3.5 Auto-Negotiation 13

5 Abbreviations 15

6 Contact 16

Tables

Table 1-1 History of the document 2
Table 1-2 Reference documents 2
Table 1-1 Component history 6
Table 3-1 Hardware Overview 8
Table 4-1 Supported AUTOSAR Standard Conform Features 9
Table 4-2 Supported PHY Test Modes 10
Table 4-3 Supported PHY Loopback Modes 10
Table 4-4 Supported PHY Tx Modes 10
Table 4-5 Supported PHY Hardware Revisions 11
Table 5-1 Abbreviations 15

1 Component History

Component Version	New Features
01.00.xx	Initial version of component
02.00.xx	QM release of component
03.00.xx	Update to latest core version. Measurement of SQI in 1000Base-T1 mode is now possible.
03.01.xx	Update latest core version Implemented initialization sequence according to Marvell's Initialization Sequence guide [5] Implemented bug fixes according to Marvell's errata sheets [6] and [7]

Table 1-1 Component history

2 Introduction

This document describes the functionality, API and configuration of the Ethernet Transceiver Driver.

Supported AUTOSAR Release*:	4	
Supported Configuration Variants:	pre-compile	
Vendor ID:	ETHTRCV_30_88Q2112_VENDOR_ID	30 decimal (=Vector-Informatik, according to HIS)
Module ID:	ETHTRCV_30_88Q2112_MODULE_ID	73 decimal

* For the precise AUTOSAR Release 4.x please see the release specific documentation.

The Ethernet Transceiver Driver provides hardware independent access to control connected transceivers in a generic way. It offers the functionality to control the mode of operation of connected transceivers as well as to determine their current state, e.g. if events like link status change or bus errors happened.

The transceiver itself is a hardware device, which mainly transforms the logical I/0 signals of the Ethernet Controller to the bus compliant electrical levels, currents and timings.

The transceiver driver considered in this technical reference is designed to operate with Marvell 88Q2110 and 88Q2112 transceivers. This documentation however only covers the special aspects of the driver software covering this kind of transceivers, while all general aspects are covered by the complementary document [1].

3 Hardware Overview

Table 3-1 summarizes the supported derivatives and settings this Ethernet Transceiver Driver can be used on. It also provides information on the documentation used during development of the driver for detailed reference to hardware specific information.

Derivatives	▶ 88Q2110 ▶ 88Q2112
Compiler	Compiler independent
Hardware Manuals	▶ 88Q2110/88Q2112 Datasheet Automotive 100/1000BASE-T1 Transceiver Doc. No. MV-S110850-00 Rev. A March 23, 2017, Draft
Safety Manuals	-
Errata Sheets	-
Specifics	-
Used Registers	See EthTrcv_30_88Q2112_LL_Regs.h
Hardware features related to independence or partitioning	-
Access Mechanism	▶ SMI management interface by utilizing Eth or EthSwt ▶ Ethernet switch internal registers by utilizing EthSwt
Hardware Diagnostics	▶ 6 PHY Test modes ▶ Internal/Remote Loopback mode ▶ Cable diagnostics ▶ Signal quality measurement

Table 3-1 Hardware Overview

4 Derivative Specific Functional Description

4.1 Overview

The general functional description of the transceiver driver is part of the general technical reference [1]. This section gives an overview over which of the features are supported by the specific transceiver driver considered in this document and to what extent. The features marked with ■ in Table 4-1 are available for the Ethernet transceiver driver for 88Q2112 transceivers. If a feature is supported with limitations, it is marked with ■. The limitations are explained in section 4.2.

Supported AUTOSAR Standard Conform Features	
Initialization of the Ethernet Transceiver	■
Setting and getting the Transceiver Mode	■
Transceiver wakeup	■
Restart auto negotiation	■
Get link state	■
Get baud rate	■
Get duplex mode	■
Set test mode	■
Set loopback mode	■
Get signal quality	■
Set TX mode	■
Get cable diagnostics result	■
Get PHY identifier	■
Retrieve version info	■
Report development errors to DET	■
Report production errors to DEM	■

Table 4-1 Supported AUTOSAR Standard Conform Features

4.2 Limitations to supported Features and Functions

4.2.1 PHY Test Modes

The PHY Test modes are explained in detail in the general technical reference [1]. In Table 4-2 all Test modes which are supported by the Ethernet transceiver driver considered in this document are marked with ■. The ones that are not supported respectively with ■.

TestMode value	
ETHTRCV_PHYTESTMODE_NONE	■
ETHTRCV_PHYTESTMODE_1	■
ETHTRCV_PHYTESTMODE_2	■
ETHTRCV_PHYTESTMODE_3	■

ETHTRCV_PHYTESTMODE_4	■
ETHTRCV_PHYTESTMODE_5	■
ETHTRCV_PHYTESTMODE_6	■
ETHTRCV_PHYTESTMODE_7	■

Table 4-2 Supported PHY Test Modes

*) The additional test modes 6 and 7 are only supported if the transceiver is running in 1000BASE-T1 mode.

4.2.2 PHY Loopback Modes

The PHY Loopback modes are explained in detail in the general technical reference [1]. In Table 4-3 all Loopback modes which are supported by the Ethernet transceiver driver considered in this document are marked with ■. The ones that are not supported respectively with ■.

LoopbackMode value	
ETHTRCV_PHYLOOPBACK_NONE	■
ETHTRCV_PHYLOOPBACK_INTERNAL	■
ETHTRCV_PHYLOOPBACK_EXTERNAL	■
ETHTRCV_PHYLOOPBACK_REMOTE	■

Table 4-3 Supported PHY Loopback Modes

4.2.3 PHY TX Modes

The TX modes are explained in detail in the general technical reference [1]. In Table 4-4 all TX modes which are supported by the Ethernet transceiver driver considered in this document are marked with ■. The ones that are not supported respectively with ■.

TxMode value	
ETHTRCV_PHYTXMODE_NORMAL	■
ETHTRCV_PHYTXMODE_TX_OFF	■
ETHTRCV_PHYTXMODE_SCRAMBLER_OFF	■

Table 4-4 Supported PHY Tx Modes

4.2.4 PHY Cable Diagnostics

The Cable Diagnostics are explained in detail in the general technical reference [1].

The 88Q2112 transceiver provides valid Cable Diagnostic results in master mode without any additional set-up. In slave mode the 88Q2112 transceiver should be detached from the master interface to provide valid Cable Diagnostic result. A short can only be detected between the positive and negative MDI signal wires. Detection of a short to ground or power supply is not supported by the 88Q2112 transceiver.

4.3 Marvell Specific Implementation

Due to specialties of the different hardware revisions and special handlings not provided in the standard datasheet [2] of Marvell 88Q2110/88Q2112 transceivers, a couple of features had to be implemented additionally according to errata sheets and sequences provided by Marvell. These specialties are described in the following sub sections.

4.3.1 Supported hardware revisions

Table 4-5 gives an overview of the supported 88Q2110/88Q2112 hardware revisions by the driver.

- Supported
- Not Support

Hardware Revision / PHY	A2	A1	A0	Z1	Z0
88Q2110	■	■	■	■	■
88Q2112	■	■	■	■	■

Table 4-5 Supported PHY Hardware Revisions

4.3.2 Initialization Sequence

Marvell 88Q2110/88Q2112 transceivers require a special initialization sequence (see [5]) for a proper Link-Up and operation in 100 Base-T1, 1000 Base-T1 or Auto-Negotiation mode.

As the initialization sequence provided by Marvell requires certain delays during the initialization the link handling main function is used to handle these delays. The calling period of the link handling main function is configurable and defines the minimal possible delay before the link handling main function is called again (see Figure 4-1).

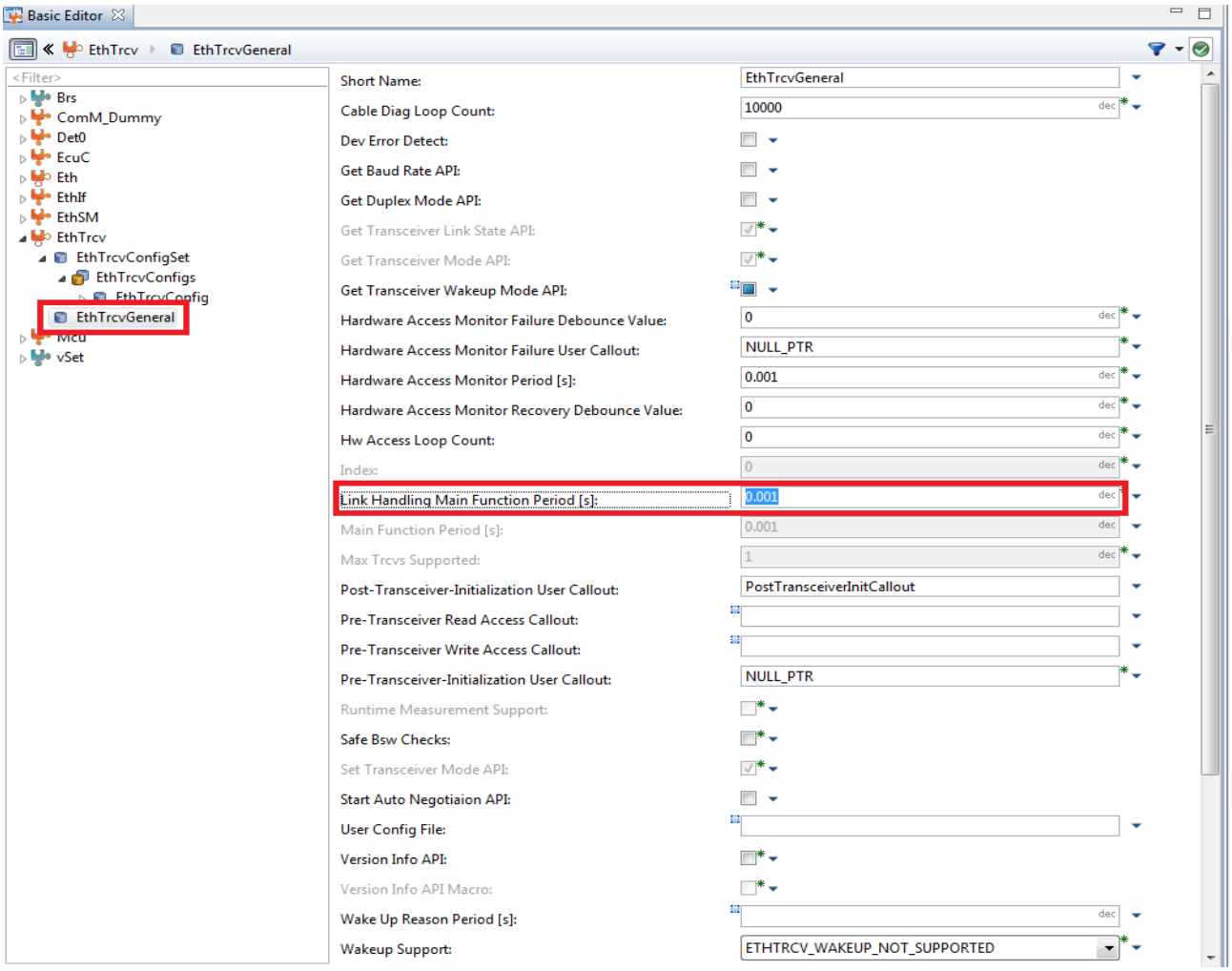


Figure 4-1 Link Handling Main Function Period

Note
It is recommended to configure a Link Handling Main Function Period of 1 millisecond.

Caution
Due to the delays required by Marvell's initialization sequence Link-Up times > 200 milliseconds are possible.

4.3.3 1000Base-T1 Mode SQI

The algorithm used to calculate the 1000Base-T1 Mode SQI value in the 88Q2112 transceiver driver was provided by Marvell. For detailed information about the algorithm see [3].

Note
The 1000Base-T1 Mode SQI algorithm is not provided in [2]. Only information about the 100Base-T1 Mode SQI calculation is available there.

4.3.4 PAM2/PAM3 Link Training

For hardware revisions before revision A1 the link training was not compliant with the IEEE specification. 88Q2110/88Q2112 hardware revisions A2 and A1 therefore provide the ability to be used in a legacy mode in order to allow link training to be completed with link-partners of previous 88Q2110/88Q2112 hardware revisions before A1.

Note
The driver does not support the functionality of 88Q2110/88Q2112 revisions A2 and A1 to be used in legacy mode.

Post-Transceiver-Init-User-Callout and Marvell's errata sheets ([6] and [8]) should be used to enable the use of legacy mode.

4.3.5 Auto-Negotiation

The transceiver can be forced to negotiate to be in master mode. In order to force the transceiver to negotiate to be the link master, Auto-Negotiation must be activated in the configuration and the Connection Negotiation must be set to ETHTRCV_CONN_NEG_MASTER (see Figure 4-2).

If Auto-Negotiation is activated and Connection Negotiation is set to ETHTRCV_CONN_NEG_AUTO the transceiver will negotiate either to be master or slave.

Forcing the transceiver to negotiate to be slave is not possible.

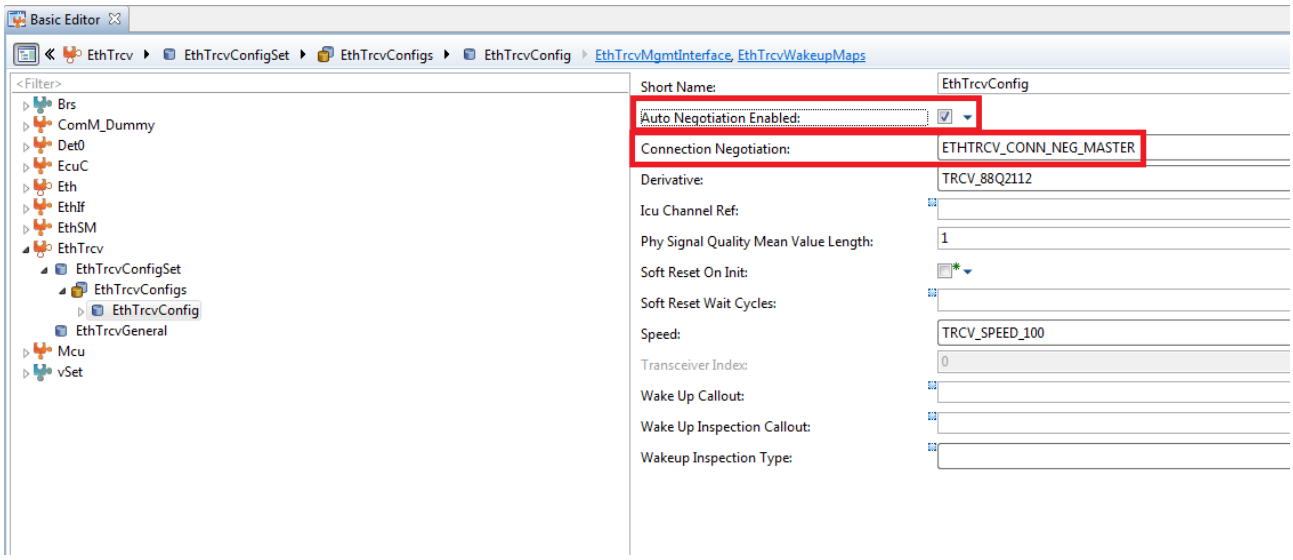


Figure 4-2 Auto-Negotiation force master

Caution

If Auto-Negotiation mode is activated a link can only be established with a link partner if the link partner also performs its Auto-Negotiation routine. For more information regarding this see [5].

5 Abbreviations

Abbreviation	Description
API	Application Programming Interface
AUTOSAR	Automotive Open System Architecture
BSW	Basis Software
DEM	Diagnostic Event Manager
DET	Development Error Tracer
Eth	Ethernet Controller Driver
EthIf	Ethernet Interface
EthSwt	Ethernet Switch Driver
EthTrcv	Ethernet Transceiver Driver
HIS	Hersteller Initiative Software
ICU	Input Capture Unit
MICROSAR	Microcontroller Open System Architecture (the Vector AUTOSAR solution)
MII	Media Independent Interface
Platform	Hardware including host and communication controller (might also be integrated in host) on which the communication stack is implemented.
SMI	Serial Management Interface
SQI	Signal Quality Index
SWS	Software Specification

Table 5-1 Abbreviations

6 Contact

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